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GitHub Link	https://github.com/192425245/ml0403.git

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 2

Industry Problem Solving Task

COURSE NAME: Deep Learning
SLOT :

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10%
Assessment Tool Weightage: 40%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I K.Neeha(192425245)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

K.Neeha

Signature of the student:

Name of the student:K.Neeha

Criteria	Understanding of Industry Problem (2)	Application of Engineering Concepts (2.5)	Solution Design (2.5)	Use of Tools (2)	Analysis (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100%
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Q1. Transfer Learning for Skin Cancer Detection

Problem Understanding

A healthcare startup wants to detect skin cancer from dermoscopic images. Since only a limited number of labeled images are available, training a deep CNN from scratch may lead to overfitting. Therefore, transfer learning is used.

Proposed Model Architecture

Input Image → Pre-trained ResNet50 → Global Average Pooling → Dense Layer → Softmax Output

Choice of Model

ResNet50 is selected because:

- It is pre-trained on ImageNet.
- Uses residual connections to avoid vanishing gradients.
- Extracts powerful image features.
- Performs well on medical image classification tasks.

Frozen Layers

- Initial Convolution Layer
- Batch Normalization Layers
- First Three Residual Blocks

These layers capture general image features such as edges, shapes, and textures.

Fine-Tuned Layers

- Last Residual Block
- Newly Added Dense Layers

Training Strategy

1. Resize dermoscopic images.
2. Apply image augmentation.
3. Load pre-trained ResNet50 weights.
4. Freeze early layers.
5. Replace final classification layer.
6. Fine-tune deeper layers using a low learning rate.

Performance Improvement

- Reduces overfitting.
- Requires less training data.
- Improves classification accuracy.

- Reduces computational cost.

Conclusion

Transfer learning with ResNet50 provides an efficient and accurate solution for skin cancer detection using limited medical images.

Q2. DenseNet and PixelNet for Road-Scene Segmentation

Problem Understanding

The smart city project requires pixel-level classification of:

- Roads
- Pedestrians
- Vehicles
- Traffic Signs

Proposed Architecture

Segmented Output Image

Role of DenseNet

DenseNet provides feature extraction through dense connections.

Dense Connections

Each layer receives outputs from all previous layers.

Benefits:

- Feature reuse
- Better gradient flow
- Reduced parameters
- Improved learning efficiency

Role of PixelNet

PixelNet performs pixel-wise prediction by combining:

- Low-level features
- Mid-level features
- High-level semantic features

Segmentation Process

1. Extract multi-scale features.
2. Combine feature maps.
3. Generate pixel-level classifications.
4. Assign labels to each pixel.

Advantages

- High segmentation accuracy.

- Better object boundary detection.
- Efficient memory utilization.
- Reduced computational complexity.

Conclusion

DenseNet + PixelNet creates a robust semantic segmentation system suitable for intelligent transportation applications.

Q3. Encoder–Decoder using Bidirectional RNN for Multilingual Chatbot

Problem Understanding

The chatbot must translate customer queries while preserving contextual meaning across different languages.

Proposed Architecture

Translated Sentence

Encoder Function

The encoder reads the source sentence.

Forward RNN:

- Reads left to right.

Backward RNN:

- Reads right to left.

Outputs are combined to create a context vector.

Context Vector

Stores:

- Sentence meaning
- Word relationships
- Context information

Decoder Function

The decoder generates translated words one at a time.

Why Bidirectional Processing?

Example:

"The bank is near the river."

The meaning of "bank" depends on surrounding words.

Bidirectional RNN considers:

- Previous words
- Future words

This improves contextual understanding.

Benefits

- Better translation quality.
- Improved context preservation.
- Handles ambiguous words effectively.

Conclusion

The Encoder–Decoder architecture with Bidirectional RNNs produces more accurate multilingual translations.

Q4. LSTM and BPTT for Stock Price Prediction

Problem Understanding

The company wants to predict future stock prices using historical market data.

Input Features

- Opening Price
- Closing Price
- High Price
- Low Price
- Trading Volume

Proposed Architecture

Why LSTM?

Traditional RNN Problems:

- Vanishing gradients
- Difficulty learning long-term dependencies

LSTM Advantages:

- Memory cells
- Forget gate
- Input gate
- Output gate

These components retain important information over long periods.

Role of BPTT

Backpropagation Through Time (BPTT) trains the network by:

1. Unrolling LSTM through multiple time steps.
2. Computing prediction errors.
3. Propagating errors backward through time.
4. Updating network weights using gradient descent.

Benefits

- Learns long-term market trends.
- Improves prediction stability.
- Handles sequential financial data effectively.

Conclusion

LSTM trained with BPTT provides accurate stock forecasting and outperforms traditional RNN models.

Q5. Integrated Deep Learning Solution for Video Caption Generation

Problem Understanding

The media streaming company wants to automatically generate captions for uploaded videos.

Proposed Architecture

Caption Generation

Component 1: Transfer Learning

Role

A pre-trained CNN extracts visual information from video frames.

Extracted Features:

- Objects
- Actions
- Scene Information

Advantages

- Faster training
- Better visual representation
- Reduced data requirements

Component 2: Encoder

The encoder receives extracted frame features and converts them into a compact representation.

Component 3: Context Vector

Stores:

- Visual content
- Temporal relationships
- Semantic information

Component 4: LSTM Decoder

The decoder generates words sequentially.

Example:

Input

Person riding a bicycle

Generated "A person is riding a bicycle on the road."

Output Layer

Softmax predicts the most probable next word at every step.

Training Process

1. Extract video frames.
2. Generate CNN feature vectors.
3. Train Encoder–Decoder network.
4. Use teacher forcing during training.
5. Optimize using cross-entropy loss.

Benefits

- Captures visual and temporal information.
- Produces meaningful captions.
- Improves accessibility for users.

Conclusion

The integrated CNN–LSTM Encoder–Decoder framework effectively generates accurate and context-aware captions for video content.

Final Conclusion

The proposed solutions apply advanced deep learning techniques such as Transfer Learning, DenseNet, PixelNet, Bidirectional RNN, Encoder–Decoder Networks, LSTM, and BPTT to solve real-world problems in healthcare, smart cities, multilingual communication, financial forecasting, and video captioning. These architectures improve feature extraction, contextual understanding, prediction accuracy, and computational efficiency, making them highly suitable for modern AI applications.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis